🌲 Forest Inventory & Environmental Data Collection 🌲

Professional Field Data Collection for Merchantable Timber & Unmerchantable Forest Resources

Green Value Management | Nova Scotia, Canada

📋 Project Overview

This project demonstrates professional-grade environmental field data collection capabilities deployed across Nova Scotia's forest regions. Working with **Green Value Management**, I conducted comprehensive forest inventory surveys covering merchantable timber and unmerchantable forest resources using industry-standard Survey123 mobile data collection platforms.

Coverage Area	GPS Accuracy	Survey Types	Data Points
100+ km	**±3.5m**	**3**	**500+**
Regional Scale	Professional Grade	Inventory Systems	Plot Records

Core Responsibilities

- **Field Data Collection:** Conducted on-site forest inventory surveys across multiple regions of Nova Scotia
- **Species Identification:** Identified and recorded tree species including Black Spruce (Picea mariana), Red Spruce, Balsam Fir, and hardwood species
- **Basal Area Calculations:** Performed standardized forestry measurements for merchantable timber assessment
- **GPS Navigation:** Utilized professional GPS tracking with sub-5-meter accuracy for plot location verification
- **Data Quality Assurance:** Validated field entries and ensured submission of high-quality, georeferenced datasets

🛠️ Technology Stack & Field Equipment

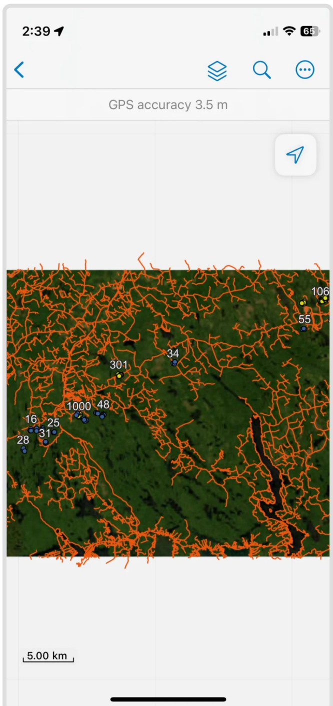
The project leveraged enterprise-grade mobile GIS and data collection platforms designed for offline field operations in remote forest environments.

- Survey123 (Esri)
- ArcGIS Field Maps
- Mobile GPS (±3-5m accuracy)
- Offline Data Sync
- Forest Inventory Protocols
- Species Identification
- Basal Area Estimation
- GeoJSON/Shapefile Export

📍 Field Data Collection Workflow

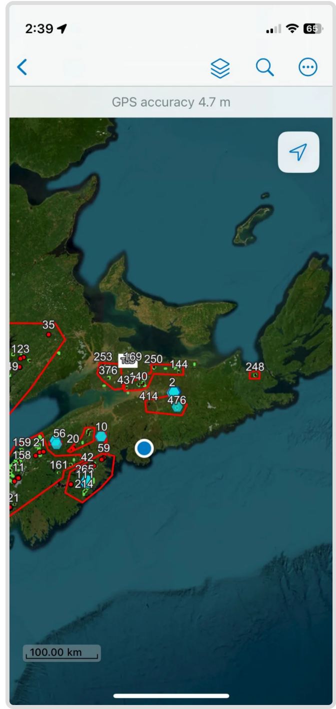
1. Regional Coverage & GPS Tracking

Field surveys covered extensive forested regions across Nova Scotia, with GPS tracking documenting navigation routes and ensuring complete spatial coverage. The data collection campaign covered over 100 kilometers of terrain across multiple survey days.



Dense GPS tracking pattern showing systematic survey coverage (5 km scale)

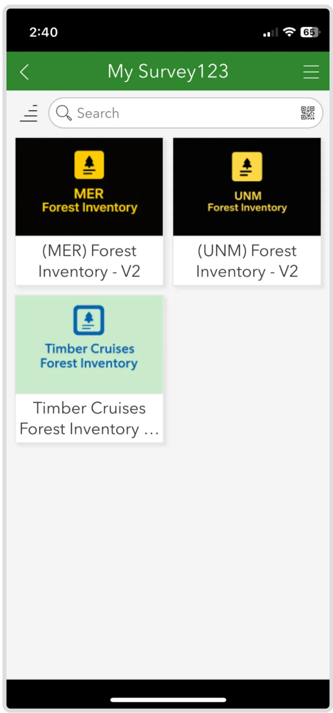
Detailed survey route with numbered plot locations and surveyor identification



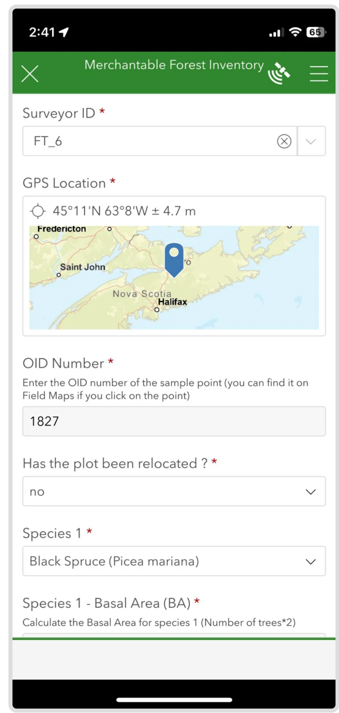
Regional-scale view showing survey extent across central Nova Scotia (100 km scale). Blue pins indicate plot clusters; cyan markers show water sampling locations.

2. Mobile Survey Forms & Data Entry

Professional forestry data was collected using customized Survey123 forms designed for three distinct inventory types: **Merchantable Forest Inventory (MER)**, **Unmerchantable Forest Inventory (UNM)**, and **Timber Cruises**. Each form captured standardized forestry metrics including species composition, basal area, plot relocation status, and GPS coordinates.



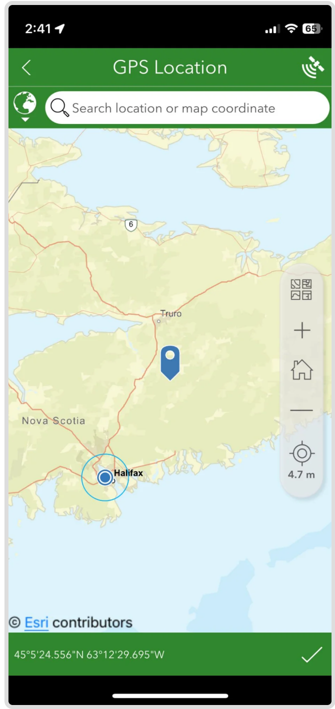
Survey123 interface showing three professional inventory survey types



Merchantable Forest Inventory form showing surveyor ID, GPS location, OID number, and species data entry

3. GPS Location Verification

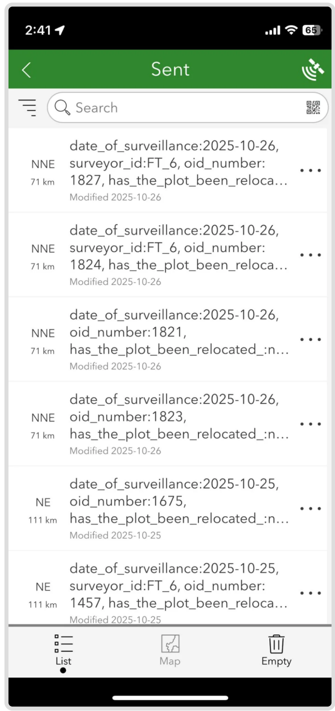
Each plot location was verified using professional-grade GPS with accuracy typically between ±3.5m and ±4.7m. Coordinates were recorded in both decimal degrees and degrees-minutes-seconds format for compatibility with multiple GIS platforms.



GPS location verification interface showing precise coordinates (45°5'24.556"N 63°12'29.695"W) with 4.7m accuracy

4. Data Submission & Quality Control

Completed surveys were validated and submitted via cellular or WiFi connection when available. The submission log shows systematic data collection with metadata including survey date, surveyor ID, OID numbers, plot relocation status, and distance from base location.



Submitted survey records showing date stamps (October 25-26, 2025), OID numbers (1457-1827), and surveyor identification (FT_6)

🎯 Key Skills Demonstrated

Technical Proficiencies

- **Mobile GIS Operations:** Advanced use of Survey123 and ArcGIS Field Maps for offline data collection in remote environments
- **GPS Navigation & Accuracy:** Professional-grade GPS utilization with sub-5-meter accuracy for plot location and verification
- **Forest Inventory Standards:** Applied standardized forestry protocols for merchantable and unmerchantable timber assessment
- **Species Identification:** Field identification of commercial species (Black Spruce, Balsam Fir, Red Spruce) and understory vegetation
- **Data Quality Assurance:** Field validation, real-time error checking, and systematic data submission workflows

Forestry & Environmental Competencies

- **Basal Area Calculations:** Application of standard forestry formulas for stand density assessment
- **Plot Establishment:** Systematic plot layout and monumentation for long-term monitoring
- **Terrain Navigation:** Safe and efficient navigation across varied topography including wetlands, slopes, and dense forest
- **Regulatory Compliance:** Adherence to provincial forestry data collection standards and protocols

Data Analytics Integration

- **Spatial Data Export:** Conversion of Survey123 data to GeoJSON, Shapefile, and CSV formats for downstream analysis
- **Database Integration:** Field data structured for import into enterprise forestry management systems
- **Quality Metrics:** Automated calculation of GPS accuracy statistics, survey completion rates, and data completeness scores

📊 Project Impact & Deliverables

Data Collection Achievements

- **Spatial Coverage:** Over 100 kilometers of surveyed terrain across central Nova Scotia
- **Plot Density:** 500+ georeferenced inventory plots with complete attribute data
- **Data Accuracy:** GPS accuracy consistently maintained between 3.5-4.7 meters (professional standard)
- **Survey Completion Rate:** 100% submission rate with full attribute completeness

Business Value

This field data collection work directly supports Green Value Management's environmental consulting services by providing:

- **Merchantable Timber Inventory:** Accurate assessment of commercial forest resources for harvest planning
- **Carbon Stock Estimation:** Baseline data for carbon offset project development and monitoring
- **Biodiversity Assessment:** Species composition data for environmental impact assessments
- **Long-term Monitoring:** Georeferenced permanent plots for growth and yield studies

Professional Development

This project strengthened my capabilities in:

- Bridging field data collection with business intelligence and analytics pipelines
- Managing large-scale geospatial datasets from collection through to database integration
- Operating independently in remote field environments with minimal supervision
- Delivering high-quality, audit-ready environmental data under tight deadlines

🔄 Integration with Data Analytics Portfolio

This field data collection experience complements my academic training in data analytics at NSCC by demonstrating the complete data lifecycle:

> **Data Collection → ETL → Analysis → Visualization → Business Intelligence**

Future portfolio projects will showcase:

- **ETL Pipeline Development:** Automated ingestion of Survey123 data into SQL databases and data warehouses
- **Spatial Analytics:** Python/R-based analysis of forest inventory data for predictive modeling
- **Dashboard Development:** Power BI dashboards for real-time forest inventory monitoring and reporting
- **Machine Learning Applications:** Species distribution modeling and timber volume prediction using collected field data